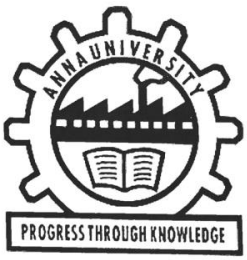
UI/UX DESIGN OF AN E-COMMERCE WEBSITE

INTERNSHIP REPORT

BACHELOR OF TECHNOLOGY
INFORMATION TECHNOLOGY

Submitted by

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JUNE 2026

UNIVERSITY COLLEGE OF ENGINEERING (BIT CAMPUS),
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DEPARTMENT OF INFORMATION TECHNOLOGY

BONAFIDE CERTIFICATE

This is to certify that the internship report entitled **"UI/UX DESIGN OF AN E-COMMERCE WEBSITE"** is a record of the internship learning and design documentation prepared by **NIRANJAN R (Register Number: 810023205002)**, B.Tech. Information Technology, University College of Engineering, BIT Campus, Anna University, Tiruchirappalli.

The report documents a UI/UX design internship at Astonish InfoTech Private Limited from 01 June 2026 to 30 June 2026. It has been prepared for academic submission. The report is not an official certificate issued by the company.

Internal Guide **Head of Department**

Date / Place: Tiruchirappalli /

Submitted for the practical examination held on _____

DECLARATION

I, NIRANJAN R (Register Number: 810023205002), declare that this report titled "UI/UX DESIGN OF AN E-COMMERCE WEBSITE" has been prepared to document the learning areas and design process associated with my UI/UX internship at Astonish InfoTech Private Limited during 01 June 2026 to 30 June 2026.

The report distinguishes the internship details stated in the supplied official documents from explanatory design assumptions and illustrative mockups. Generated interface visuals, wireframes and diagrams are proposed examples and are not represented as screenshots from an original Figma project. No functioning website, backend, payment integration, or formal usability study is claimed.

Signature of Student: _____ Date: _____

ACKNOWLEDGEMENT

I express my sincere gratitude to Astonish InfoTech Private Limited for providing the opportunity to undertake an offline internship in the UI/UX domain during June 2026. The internship documentation provided a context for studying interface design concepts and considering how these concepts can be applied to an e-commerce experience.

I thank the faculty members of the Department of Information Technology, University College of Engineering, BIT Campus, Anna University, Tiruchirappalli, for their academic guidance and support. Names of individual faculty guides have not been inserted because they were not supplied in the source documents.

I am grateful to my family and friends for their encouragement during my academic work.

ABSTRACT

This report documents a 30-day offline UI/UX Design internship at Astonish InfoTech Private Limited, Tiruchirappalli, from 1 June 2026 to 30 June 2026. The completion certificate identifies the internship topic as UI/UX Design, while the annexure specifies the project work as “E-Commerce”. Accordingly, the report adopts the academic project title “UI/UX DESIGN OF AN E-COMMERCE WEBSITE” and explores the application of interface design principles to the online shopping journey, from product discovery to purchase.

The learning schedule is divided into three phases: UI/UX fundamentals during Days 1–5, advanced design concepts during Days 6–15, and e-commerce project planning and interface documentation during Days 16–30. Key topics include usability heuristics, perception principles, colour theory, typography, spacing, grids, user-centred design, personas, user journeys, information architecture, wireframes, high-fidelity mockups, reusable components, prototyping, responsive design, accessibility and interface content. Figma is used as the primary design tool, with ShopEase introduced as a fictional e-commerce store to demonstrate a consistent proposed experience across the home page, product listing, product detail, cart, checkout and mobile screens. The report also presents proposed experience requirements, empty, error and loading states, a visual design system with calculated colour-contrast ratios, a recommended Figma file structure, prototype interactions and a design-review checklist. For example, a bright pink accent achieves approximately 4.35:1 contrast against white, falling below the 4.5:1 guideline for normal text [2], supporting the recommendation to use a darker shade for text-bearing buttons.

The report follows a clear evidence-based approach by distinguishing verified internship details from the supplied learning schedule, established design knowledge, illustrative content and unverified claims. This distinction ensures that the fictional store and proposed design solutions are not misrepresented as actual client data or completed implementation work.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
UI	User Interface
UX	User Experience
HCI	Human–Computer Interaction
IA	Information Architecture
CTA	Call to Action
RGB	Red, Green, Blue colour model
WCAG	Web Content Accessibility Guidelines
AA	WCAG conformance Level AA
Lo-fi	Low-fidelity
Hi-fi	High-fidelity
RWD	Responsive Web Design
ISO	International Organization for Standardization
PIN	Postal Index Number
px	Pixel
PDF	Portable Document Format

1. INTRODUCTION

1.1 Overview

E-commerce websites allow people to discover products, compare alternatives, review product details, and move through purchase-related steps using a digital interface. The quality of that interface strongly influences how easily visitors understand the available choices and complete their intended tasks. A visually attractive page is not sufficient by itself: content must be understandable, navigation predictable, and interactive elements consistent from one screen to the next.

User Interface (UI) design concerns the visible and interactive elements of a digital product, including layout, typography, colour, buttons, fields, cards, and menus. User Experience (UX) design considers the broader experience of using the product, including task flow, clarity, feedback, and the effort required to reach a goal. The two areas overlap and should be considered together: a well-styled button placed at the wrong point in a purchase flow still produces a poor experience, and a well-planned flow presented with unreadable text still frustrates the user.

Online shopping interfaces are particularly demanding because they combine several kinds of interaction in one journey. Shoppers browse and search, compare items, make choices about size or colour, manage a cart, enter personal and delivery information, and finally confirm a purchase. Each of these activities has its own design requirements, and the transitions between them must remain clear. This report studies how established UI/UX principles can be applied across that journey in a proposed e-commerce interface.

1.2 Internship Details

The work documented in this report relates to a one-month offline internship in the UI/UX domain at Astonish InfoTech Private Limited, Tiruchirappalli. Except where a row is marked otherwise, the details below are taken from the offer letter, completion certificate, attendance certificate and annexure supplied by the student [10], which are reproduced in Appendix A.

Table 1.1: Internship details

Item	Details
Student	NIRANJAN R
Registration number	810023205002
Degree / Department	B.Tech. / Information Technology
Institution	University College of Engineering, BIT Campus, Anna University, Tiruchirappalli
Organization	Astonish InfoTech Private Limited
Role / Domain	Intern / UI/UX

Item	Details
Mode	Offline
Period	01 June 2026 – 30 June 2026 (30 days)
Project entry in annexure	E-Commerce
Primary design tool	Figma (taken from the learning plan supplied by the student; not stated in the internship documents)

1.3 Problem Context

For a general online store, shoppers may encounter friction when categories are hard to scan, product information is incomplete, controls look different across pages, or the next step in a purchase flow is unclear. Typical symptoms include crowded product pages, ambiguous button labels, hidden filters, forms that discard entered data after an error, and cart pages that do not make totals understandable. These are common design risks rather than verified defects in a specific commercial website.

The proposed project responds to these risks by organizing a consistent and understandable shopping journey. Rather than evaluating an existing website, it uses a fictional store concept, named ShopEase, as a vehicle for explaining how structure, hierarchy, components and interaction states can be planned before development begins.

1.4 Project Objectives

The project aims to plan a clear e-commerce interface using UI/UX principles and Figma-oriented design documentation. The specific objectives are:

- To structure product categories and navigation so that shoppers can find items by browsing or searching.
- To establish a readable visual hierarchy that directs attention to the most important content on each screen.
- To present product information consistently across listing, detail and cart screens.
- To plan product listing, product detail, cart and checkout screens as one coherent purchase journey.
- To define reusable visual elements, including colour, typography, spacing and component states.
- To consider both desktop and mobile layouts and the adaptation between them.
- To document interaction concepts for a possible prototype and a checklist for later design review.

1.5 Scope and Boundaries

The scope is limited to design analysis, proposed information architecture, wireframe and mockup concepts, component planning, and prototype-flow documentation. The report does not claim implementation of a production website, frontend code, backend services, databases, authentication, inventory management, or a payment gateway. Formal user interviews, usability tests, quantitative evaluations, and client acceptance are not documented because supporting evidence was not supplied.

The illustrations included later in this report are generated explanatory mockups, wireframes and diagrams. They are not authenticated captures from the student's Figma workspace. Where a figure or table describes a proposed design, the caption or surrounding text says so explicitly.

1.6 Significance of the Study

Interface quality has practical consequences. Clear labels reduce the number of mistakes a shopper makes; consistent components reduce the effort of learning a site; visible feedback reduces uncertainty about whether an action has succeeded; and accessible design widens the range of people who can use a service. For a development team, a documented design also reduces ambiguity at handoff, because the intended behaviour of each screen and component is described before engineering effort is spent.

1.7 Method of Documentation and Evidence Classification

The report was prepared from four kinds of material: the internship documents supplied by the organization, the 30-day learning schedule provided by the student, widely taught UI/UX concepts, and illustrative visuals created to explain the concepts. To keep the report truthful, every statement is treated as belonging to one of the categories in Table 1.2. This classification is applied throughout the document and is summarized again in the conclusion.

Table 1.2: Evidence categories used in this report

Category	Meaning	Examples in this report
Source-supported	Stated in the supplied internship documents	Organization name, dates, domain, mode, annexure project title
Learning schedule	Provided by the student as the planned learning topics	Day-wise topics in Chapter 6
Explanatory	General UI/UX knowledge used to explain concepts	Heuristics, Gestalt principles, WCAG, responsive design
Illustrative	Generated or drawn visuals and fictional content	ShopEase mockups, wireframes, diagrams, sample palette
Not claimed	Items for which no evidence was supplied	Live website, Figma file link, usability test results, client approval

1.8 Report Organization

Chapter 2 records the organization and internship context based on the supplied documents. Chapter 3 explains foundational UI/UX principles, including heuristics, perception, colour, typography and layout. Chapter 4 develops user-centred design, information architecture, wireframing, prototyping, responsive design, accessibility and content. Chapter 5 documents the proposed e-commerce interface screen by screen, together with a visual design system, a Figma file structure and a prototype plan. Chapter 6 summarizes the 30-day learning schedule and the outcomes. Chapter 7 presents the conclusion, limitations and future enhancements. References and the original supporting documents (Appendix A) follow.

2. ORGANIZATION PROFILE AND INTERNSHIP OVERVIEW

2.1 Organization Details

The supplied internship offer letter identifies Astonish InfoTech Private Limited in Tiruchirappalli [10]. The letterhead lists G7, Technology Park, Bharathidasan University (BUTP), Kajamalai Campus, Tiruchirappalli – 620 023. It also lists the email admin@astonishinfotech.com, website www.astonishinfotech.com, and office telephone numbers 0431-2331318 and 09994954031. These details are reproduced from the supplied documents and are not independently verified here.

Table 2.1: Organization details supported by the supplied documents

Field	Source-supported detail
Organization	Astonish InfoTech Private Limited
Location	G7, Technology Park, Bharathidasan University (BUTP), Kajamalai Campus, Tiruchirappalli – 620 023
Internship mode	Offline
Domain	UI/UX
Offer letter date	30.05.2026
Internship period	01/06/2026 to 30/06/2026
Annexure project title	E-Commerce
Industrial expert / joint supervisor	Srikanniga Priyadharshini R (as handwritten in annexure)
Qualification (annexure)	B.Tech – IT (Hons) (as handwritten in annexure)
Designation	Project Manager (as handwritten in annexure)

2.2 Services Listed on the Organization Letterhead

The letterhead footer of the offer letter and annexure lists the organization's service areas as web design, web development, web hosting, domain registration, digital marketing and Android. This listing is useful context for the internship because UI/UX design sits between several of these areas. Web

design and web development both depend on interface decisions made before coding; digital marketing relies on landing pages and product pages that communicate clearly; and Android work raises the same layout and interaction questions on a smaller screen.

2.3 Internship Overview

The offer letter provides an internship opportunity to Niranjan R, registration number 810023205002, in the UI/UX domain for the period 1 June 2026 to 30 June 2026, with the offer dated 30 May 2026 and the mode recorded as offline [10]. The completion certificate states that the student completed an internship in the topic entitled UI / UX Design during the same dates, and carries the seals and signatures of an authorised signatory and a mentor. The attendance certificate records the same student, programme, department and period, and shows the semester as VII.

The annexure lists “E-Commerce” as the title of the project work and records the duration of the internship as one month. This report uses the more descriptive academic title “UI/UX DESIGN OF AN E-COMMERCE WEBSITE” to align the report with the project requirement of the department while retaining the source wording when discussing the certificate. The annexure heading refers to an internship in June–July 2026, whereas the offer letter, completion certificate and attendance certificate all give 1 June to 30 June 2026; this report follows the dates stated in those three documents.

2.4 Supervision and Documentation Requirements

The annexure lists terms and conditions that frame the internship: the student is to meet the institute supervisor periodically and attend review committee meetings for evaluating progress; the final report of the internship, duly signed by the industrial expert, is to be submitted to the college; and an attendance certificate in the prescribed format, signed by the competent authority of the industry, is to be produced. This report is prepared in that academic context. The signature fields on the bonafide certificate page are intentionally left for the department and the industrial expert to complete.

2.5 Learning Context

An internship provides an opportunity to connect classroom concepts with a practical domain. In UI/UX design, this connection includes converting broad user needs into organized page structures, comparing alternative layouts, applying visual consistency, and explaining why an interaction should be clear to the user. The report organizes the supplied learning categories into a progressive sequence rather than claiming unsupported company-specific events.

The sequence moves from vocabulary to method to application. Fundamentals give the designer a language for describing interfaces; advanced concepts

supply methods for planning and checking them; and the e-commerce project applies both to a concrete screen set. This progression also reflects the way design work is normally staged: understand the problem, structure the solution, refine its appearance, and review it against criteria.

3. UI/UX DESIGN FUNDAMENTALS

3.1 Understanding UI and UX

UI design focuses on the interface that people see and operate. Its concerns include typography, colours, layout, component appearance, spacing, iconography, and visual states. UX design considers the complete interaction with a product: whether users can identify what to do, understand the result of an action, recover from mistakes, and accomplish their goals without unnecessary effort [5], [8].

For an e-commerce site, UI decisions include the style of a product card, the contrast of an "Add to Cart" button, and the arrangement of price and product name. UX decisions include whether shoppers can filter a category, compare details, edit quantities in the cart, and understand the checkout sequence. A strong design coordinates both dimensions.

Table 3.1: UI and UX comparison

Aspect	UI emphasis	UX emphasis
Primary question	How does the interface look and respond?	How well does the experience support user goals?
Examples	Colour, typography, spacing, buttons	Navigation, task flow, feedback, error recovery
E-commerce application	Consistent product cards and controls	Find, evaluate, and purchase products clearly
Evaluation approach	Visual consistency and legibility review	Task-based usability review, when performed
Typical artefacts	Style guide, component library, mockups	Journey map, user flow, wireframes, test plan

3.2 Principles of Effective Design

Clarity means that labels and controls communicate their purpose. Consistency means that similar actions use similar visual treatments and behaviour. Hierarchy guides attention through size, weight, contrast, position, and spacing. Feedback communicates that an action has been recognized, such as a cart count changing after a product is added. Error prevention helps users avoid avoidable mistakes; error recovery gives them a clear way to correct an issue [5], [6].

These principles are especially important in shopping interfaces because the user may move repeatedly between category lists, product details, cart contents, and account or checkout forms. A design system should reduce the

need to relearn controls from screen to screen. Additional principles that support the same aims include simplicity (showing only what is needed at each step), proximity of related items, predictability of outcomes, and tolerance for user error.

3.3 Usability Heuristics Applied to E-Commerce

Usability heuristics are broad rules of thumb used to inspect an interface for problems. The ten heuristics published by Jakob Nielsen and Nielsen Norman Group [11] are widely used in design reviews [4]. Table 3.2 summarizes each heuristic and shows how it might be applied to the proposed shopping interface. The table is explanatory; it does not record the results of a heuristic evaluation.

Table 3.2: Usability heuristics [11] and possible e-commerce applications

Heuristic	Possible application in a shopping interface
Visibility of system status	Cart count updates after adding an item; checkout shows the current step; loading states are visible.
Match between system and real world	Use familiar words such as “Cart”, “Delivery” and “Returns” instead of internal jargon.
User control and freedom	Allow quantity edits, item removal, back navigation and a way to leave checkout without losing the cart.
Consistency and standards	Same primary button style, price format and card layout on every listing.
Error prevention	Disable unavailable sizes, flag missing required options before adding to cart, validate fields as the user moves on.
Recognition rather than recall	Show selected filters, cart summary and chosen options on later screens so users do not have to remember them.
Flexibility and efficiency of use	Search for experienced users, category browsing for exploratory users, saved addresses for returning users where supported.
Aesthetic and minimalist design	Group secondary details in tabs or accordions; keep one dominant call to action per screen.
Help users recognize, diagnose and recover from errors	Plain-language messages that name the field and the fix, such as “Enter a 6-digit PIN code”.
Help and documentation	Brief size guidance, delivery and returns information, and a visible support route.

3.4 Gestalt Principles and Visual Perception

Gestalt principles describe how people tend to perceive visual groupings. They are useful because a designer can use them to make the structure of a page evident without adding extra lines or labels. In a product listing, for example, the repeated card shape and equal spacing allow the eye to read a grid as a set of comparable items even before any text is read.

3.5 Colour Theory and Palette Selection

Colour supports hierarchy, meaning, and brand expression, but colour alone should not carry essential information. A restrained palette can use a primary colour for major actions, neutral colours for backgrounds and text, and a distinct accent for secondary emphasis. Status colours should be used carefully and consistently. Contrast must be considered for text, icons, button labels, and focus indicators [2].

The illustrative design uses navy and blue for structure, a light neutral background for content areas, and a warm accent for selected calls to action. This is a proposed palette for demonstrating consistency, not an official Astonish InfoTech brand palette. Colour values should be validated against accessibility contrast guidance before implementation.



Figure 3.1: Illustrative visual direction for an e-commerce interface; not an original Figma screenshot.

To make the palette discussion concrete, the contrast ratios in Figure 3.2 were calculated using the relative-luminance formula defined in the Web Content Accessibility Guidelines [2]. WCAG 2.2 sets a minimum of 4.5:1 for normal-size text and 3:1 for large text and for meaningful user-interface components at Level AA [2]. The calculation is repeatable and is not a formal accessibility audit of any product.

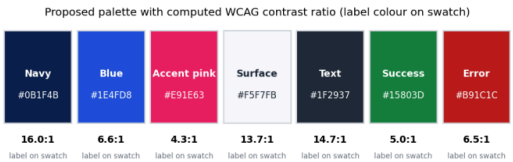


Figure 3.2: Proposed palette with calculated contrast ratios of the label colour against each swatch.

Table 3.3: Calculated contrast for the proposed palette (WCAG 2.2 AA thresholds [2])

Swatch	Background	Label colour	Ratio	Result (WCAG 2.2 AA)
Navy	#0B1F4B	#FFFFFF	16.01:1	Meets 4.5:1 for normal text

Swatch	Background	Label colour	Ratio	Result (WCAG 2.2 AA)
Blue	#1E4FD8	#FFFFFF	6.63:1	Meets 4.5:1 for normal text
Accent pink	#E91E63	#FFFFFF	4.35:1	Large text and UI components only
Surface	#F5F7FB	#1F2937	13.69:1	Meets 4.5:1 for normal text
Text	#1F2937	#FFFFFF	14.68:1	Meets 4.5:1 for normal text
Success	#15803D	#FFFFFF	5.02:1	Meets 4.5:1 for normal text
Error	#B91C1C	#FFFFFF	6.47:1	Meets 4.5:1 for normal text

The calculation reveals a useful design point. The bright pink accent visible in the illustrative mockups produces a ratio of about 4.35:1 against white, which falls just below the 4.5:1 threshold for normal-size button text, although it satisfies the 3:1 threshold for large text and for the button boundary [2]. A slightly deeper pink such as #BE185D reaches about 6.0:1 against white. The proposed design system in Chapter 5 therefore recommends the deeper value for text-bearing buttons. This is an example of how a design decision can be supported by a measurable check rather than by appearance alone.

3.6 Typography and Readability

Typography establishes information hierarchy. Page titles should be visually distinct from section headings, product names, prices, supporting descriptions, and metadata. A small set of type sizes and weights is easier to maintain than a different style for every element. Line length, line spacing, and contrast also affect readability; body text is generally easiest to read at moderate line lengths and with generous line spacing.

Product pages need particularly clear presentation of the product name, price, key attributes, delivery or returns information when available, and primary action. Dense paragraphs should be broken into meaningful sections. Text should remain readable at common viewport sizes and browser zoom levels, and layouts should tolerate larger text settings without overlap.



Figure 3.3: Proposed type scale for the illustrative interface (sizes in pixels).

3.7 Spacing, Alignment and Visual Hierarchy

Spacing creates relationships between elements. Related information should

be grouped, while unrelated sections should have enough separation to avoid visual confusion. Alignment gives a page structure: product cards, form fields, headings, and action buttons should follow a consistent grid. Repeated spacing tokens help maintain rhythm across different screens.

Visual hierarchy directs attention to the most important content first. In a product listing, the category title and filters help users orient themselves; product imagery and names support scanning; price and key information support comparison; and the action provides a clear next step. A page should avoid competing calls to action that make the next action ambiguous.

3.8 Wireframing

A wireframe is a simplified representation of a screen's structure. Low-fidelity wireframes emphasize content blocks, page hierarchy, and flow rather than final colours or imagery. They make it easier to discuss whether a page includes the necessary information before spending time polishing visual details [8].

For an e-commerce home page, a wireframe may reserve areas for the header, search, category navigation, promotional content, featured categories, product grid, and footer. For a product detail page, it may allocate space for images, product name, price, attributes, quantity, primary action, and supporting details. These are proposed layouts for this report and are presented in Chapter 5.

3.9 Introduction to Figma

Figma is a collaborative interface design tool commonly used for frames, components, styles, and interactive prototypes [1]. A design file can organize related screens into pages and sections. Components can support consistency by allowing repeated controls to share a common base design. Prototypes connect screens or states to demonstrate an intended interaction sequence.

Relevant features include frames, auto layout, constraints, shared styles or variables, and component variants [1]. This report does not claim that a specific Figma file, public prototype, or component library was submitted, because no such link or project file was provided.

4. ADVANCED UI/UX DESIGN CONCEPTS

4.1 User-Centred Design

User-centred design starts from the tasks, context, and needs of intended users rather than from visual decoration alone. For a general online store, plausible tasks include finding a category, narrowing a product list, inspecting a product, choosing an option, and reviewing the cart. The design should support these tasks with understandable labels and predictable navigation.

The international standard ISO 9241-210 describes human-centred design as an

iterative process built around four activities: understanding and specifying the context of use, specifying the user requirements, producing design solutions, and evaluating the designs against the requirements [7]. The activities repeat until the design adequately meets the requirements. This report follows that structure in outline, but the evaluation activity is described as a proposal rather than as completed work.

The personas and requirements in this report are inferred design assumptions. They are not the outcome of documented interviews, surveys, analytics, or field research. Their purpose is to illustrate how a designer can turn a target audience description into testable design questions.

4.2 Assumed Personas and Needs

Table 4.1: Assumed user needs and design responses

Assumed user type	Possible goal	Design response
Time-conscious shopper	Find a relevant item quickly	Visible search, categories, filters and sorting
Comparison-focused shopper	Review options before deciding	Consistent cards, clear price and product attributes
Mobile shopper	Browse with limited screen space	Responsive layout, touch-friendly controls, concise content
Returning customer	Review cart and proceed efficiently	Persistent cart access and understandable checkout steps

4.3 User Journeys

A user journey describes the stages a person may pass through to achieve a goal. In a shopping journey, a person may arrive at the home page, search or browse a category, inspect products, open a detail page, add an item to the cart, review the order, and proceed to checkout. At each stage the interface should answer a simple question: where am I, what choices are available, and what happens next?

Journey maps can also identify uncertainty. For example, a user may need to know whether a product option is selected, whether cart contents changed, or whether an order was confirmed. Proposed interface states should make such changes visible. Figure 4.1 shows a hypothetical emotional curve for a shopper; the values are illustrative assumptions used to show how a journey map highlights risky stages such as form entry, and they are not measurements.

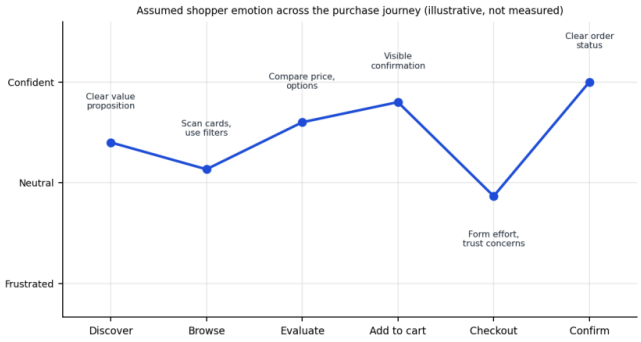


Figure 4.1: Assumed shopper emotion across the purchase journey (illustrative, not measured).

4.4 Information Architecture and Navigation

Information architecture organizes content so that users can locate it [8]. Product categories should use understandable names, and related products should be grouped consistently. A global header may contain the store identity, category navigation, search, account access, and cart access. A breadcrumb trail can help users understand where they are within a category hierarchy.

Navigation should be consistent across pages, but it must also adapt to available space. A desktop header may display several navigation items in one row; a mobile version may use a menu control and a simplified search layout. Essential actions should remain discoverable and should not depend on hover-only interaction.

Card sorting, in which participants group content labels in a way that makes sense to them, is a common method for choosing category names and structure. No card-sorting study was carried out for this report, so the proposed structure in Figure 4.2 relies on conventions familiar from online stores and would need validation with real users and a real catalogue.

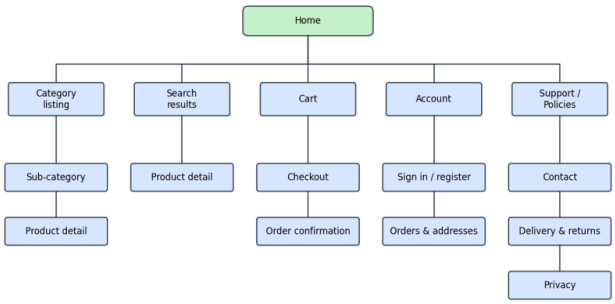


Figure 4.2: Proposed site map for the illustrative store (information architecture).

4.5 User Flows

A user flow shows the sequence of screens and decisions involved in a task. The proposed purchase flow begins at the home page or product listing, continues to product details, adds an item to the cart, and moves to checkout. If

authentication is required, the flow should provide a clear sign-in or registration branch and a way to return to the intended task.

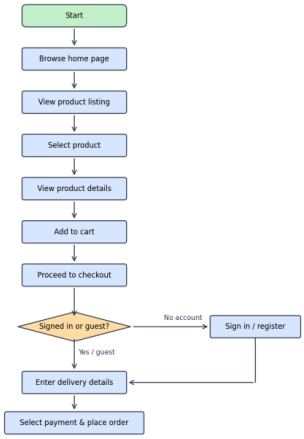


Figure 4.3: Proposed shopping flow, shown as an illustrative planning diagram.

Flow diagrams are planning tools. They do not imply that the corresponding transitions have been implemented or tested. During design review, the flow can be checked for dead ends, unclear decisions, missing confirmation, and opportunities to edit information before final submission.

A single main path hides the error and exception cases that cause many usability problems. Figure 4.4 extends the flow by showing two such branches: a shopper who tries to add a product without choosing a required option, and a shopper whose cart contains an item that has become unavailable. In each case the interface should explain the issue and offer an immediate way forward.

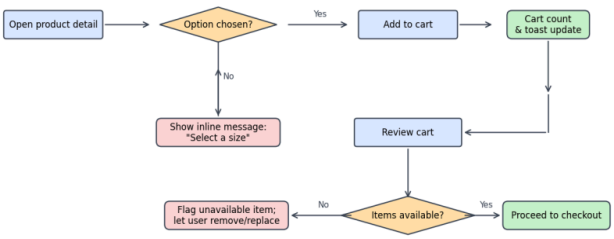


Figure 4.4: Exception branches in the add-to-cart and cart review steps (illustrative).

4.6 Low- and High-Fidelity Design

Low-fidelity designs help establish content placement and page structure with minimal visual detail. High-fidelity mockups show more realistic typography, colour, spacing, imagery, and component styling. A sensible workflow moves from structure to visual refinement so that design effort is not wasted polishing a layout that may later change.

For this project, a home page, listing page, product details page, cart, and checkout are a coherent screen set. The mockups later in the report are

examples of how these screens might be presented consistently. They are not records of the exact screens created during the internship.

4.7 Components and Design Consistency

Reusable components can represent buttons, input fields, product cards, navigation items, quantity controls, and status messages [9]. A component should have a clear purpose and a predictable set of states. Examples include default, hover, focus, disabled, loading, selected, and error states, where relevant. Not every component needs every state.

Consistency does not mean that all elements look identical; it means that elements with the same purpose behave and appear in a recognizable way. A primary action should be visually prominent across the shopping journey, while secondary actions should be less dominant. Reusable components reduce visual drift when a design is revised.

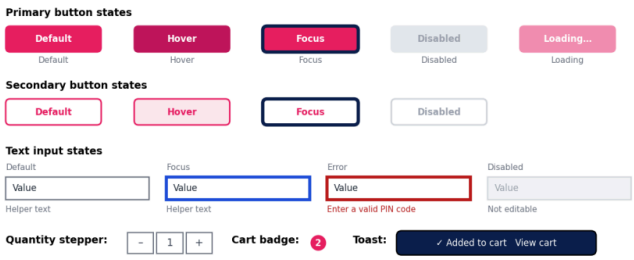


Figure 4.5: Proposed component states for buttons, inputs, quantity stepper, cart badge and toast message (illustrative).

4.8 Prototyping and Interaction Concepts

A prototype connects screens or component states to demonstrate an interaction [1]. A shopping prototype may link a product card to a product detail screen, connect an add-to-cart action to a cart state, and demonstrate the path from cart to checkout. Prototyping is useful for reviewing sequence and clarity before software development.

A prototype should not be confused with a production system. A visual transition does not establish that a database has stored a cart, that payment has been processed, or that an order has been submitted to a real service. Such behaviour requires implementation and separate verification.

4.9 Responsive Design

Responsive design adapts content and layout to different viewport widths. On desktop, a product grid may show several columns, while a mobile layout may show one or two columns depending on content and screen size. Navigation may collapse into a menu, filters may become a drawer, and product detail information may stack vertically.

Responsive adaptation should preserve content priority and interaction clarity. Touch targets need enough space, controls should not overlap, and important

information should not be hidden behind difficult interactions. Mobile layouts should be reviewed at realistic sizes rather than created by simply shrinking a desktop page.

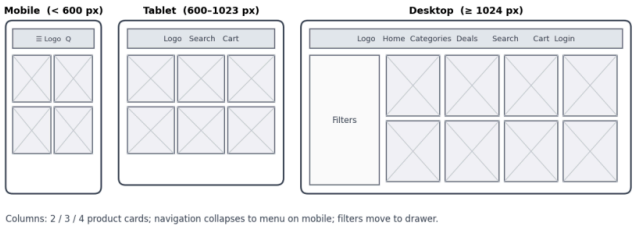


Figure 4.6: Proposed layout adaptation across mobile, tablet and desktop widths (illustrative).

4.10 Usability and Accessibility Considerations

Usability review asks whether a user can understand and complete an intended task. Useful review questions include whether labels are clear, whether the primary action is easy to identify, whether form errors explain how to recover, and whether cart changes are communicated. These questions can be used in a design inspection even when no formal usability study has taken place.

Accessibility considerations include sufficient text contrast, keyboard focus visibility, meaningful labels, understandable link text, alternatives for informative images, and avoiding colour as the sole indicator of status [2], [3]. The Web Content Accessibility Guidelines (WCAG) provide a recognized framework for accessibility requirements, organized around four principles: content should be perceivable, operable, understandable and robust [2]. This report does not claim formal WCAG conformance.

4.11 Content Design and Microcopy

Interface text is part of the design. Labels, button text, helper text and error messages determine whether an interaction feels clear or confusing. Good microcopy is specific, concise, and written in the language the user would use. Button labels should describe the outcome (“Add to cart”, “Place order”) rather than a generic action (“Submit”).

4.12 Evaluating Designs Without Formal User Studies

When real users are not available, designers can use structured inspection methods: a heuristic evaluation against the heuristics in Table 3.2 [4], [11], a cognitive walkthrough that asks at each step whether the user will know what to do and understand the feedback, and a consistency audit of components across screens. These methods find problems early but do not replace testing with real users. Section 5.19 outlines the review criteria that such an inspection could use.

5. UI/UX DESIGN OF AN E-COMMERCE WEBSITE

5.1 Project Overview

The project is a design-focused study of a proposed e-commerce website experience. Its purpose is to illustrate how a shopper can move from discovering products to reviewing a selected item and progressing through a purchase-related flow. The proposed interface is intended to be clear, consistent, and adaptable to desktop and mobile contexts.

The official annexure identifies the project work as "E-Commerce" [10]. The longer title used here describes the UI/UX focus specified for this report. The scope is interface design and documentation; no deployed website or functioning transaction system is claimed.

5.2 Objectives and Deliverables

Table 5.1: Project objectives and design deliverables

Objective	Proposed design deliverable
Improve discoverability	Home page structure, category navigation and search concept
Support product comparison	Consistent product listing cards and filter/sort concepts
Explain product choices	Product detail layout with price, options and primary action
Make purchase steps understandable	Cart and checkout screen concepts with clear progression
Maintain consistency	Reusable component and visual-style specification
Consider mobile use	Responsive layout concepts for selected screens
Document interaction sequence	User-flow diagram and prototype connection plan

5.3 Intended Users and Assumptions

The intended audience is a general online shopper using a desktop or mobile browser. Assumptions include that shoppers benefit from visible categories, search, understandable product details, consistent pricing presentation, and clear cart controls. These assumptions are reasonable starting points for design planning but should be validated with actual users before a production release.

The proposed store name "ShopEase" and all products, prices, promotional copy, and imagery shown in the illustrations are fictional placeholders created only to explain interface structure. They are not business data supplied by Astonish InfoTech or evidence of an existing commercial client.

5.4 Proposed Experience Requirements

Before screens are drawn, a designer normally writes down what the

experience must support. Table 5.2 lists proposed requirements for the illustrative store. They are written as design requirements rather than engineering specifications, and each is linked to the screen where it is addressed. Priorities use a simple scale: Must (essential to the journey), Should (important), and Could (desirable if time permits).

Table 5.2: Proposed experience requirements

ID	Requirement	Screen	Priority
R1	Shoppers can reach any top-level category from the header on every page	All	Must
R2	Search is visible on every page and accepts free text	All	Must
R3	Listing cards show image, name and price in a fixed order	Listing	Must
R4	Active filters are visible and can be cleared in one action	Listing	Must
R5	Required product options are identified before “Add to cart” succeeds	Detail	Must
R6	Cart count updates visibly after an item is added	Header / Detail	Must
R7	Cart shows option, quantity, unit price and line total for each item	Cart	Must
R8	Quantity can be changed and items removed from the cart	Cart	Must
R9	Checkout shows the current step and allows review before placing the order	Checkout	Must
R10	Form errors name the field and explain the correction	Checkout / Account	Must
R11	Layouts adapt to mobile, tablet and desktop widths	All	Must
R12	Wishlist or save-for-later control on cards and detail page	Listing / Detail	Could

5.5 Information Structure

The proposed information structure has a home page, category or product listing, product detail, cart, checkout, and order confirmation. Account access may be offered when needed. The header provides global navigation and access to search and cart. The footer can contain supporting links such as contact, delivery information, returns, and privacy information if those policies exist in the eventual product. Figure 4.2 shows the structure as a site map.

The structure aims to keep product discovery separate from transaction steps while allowing users to move between them. A shopper should be able to return to product details or adjust cart quantities without losing orientation. The final information structure would depend on the actual product catalogue and business rules, neither of which was supplied.

5.6 Home Page Design

The home page is the entry point to the proposed store. It combines a global header, search, a prominent promotional area, category shortcuts, and featured product cards. The main visual should support the purpose of the store without competing with navigation. Search and category controls should be recognizable, and the product cards should present information in a consistent order.

A home page should avoid relying exclusively on a large banner. Users who already know what they need should still be able to search or browse categories quickly. The proposed layout includes a clear hierarchy between the main promotion, category navigation, and product discovery area.



Figure 5.1: Illustrative e-commerce home page mockup (fictional ShopEase concept).

Figure 5.2 shows the corresponding low-fidelity wireframe. It makes the intended order of content explicit: header, hero with a single primary call to action, a strip of reassurance messages, category shortcuts, and a featured product grid. A designer can use this view to check whether the page answers the shopper's first questions (what is sold here, how do I find it, can I trust it) before visual styling is added.

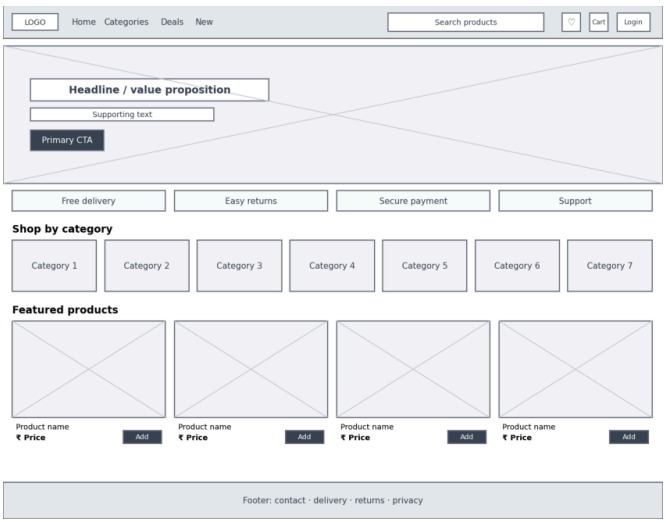


Figure 5.2: Low-fidelity wireframe of the home page (illustrative).

5.7 Product Listing Page

The product listing page supports scanning and comparison. A page title or breadcrumb identifies the current category. Filters and sorting options help narrow a larger catalogue, while product cards use a consistent image area, name, price, and optional rating or brief attribute. The design should communicate when filters are active and provide a simple way to clear them.

The number of columns should respond to viewport width. On smaller screens, filters may be placed in a collapsible panel, but the selected filter state should remain visible. Product cards should not hide essential information behind hover interactions, because touch devices do not provide the same hover behaviour.

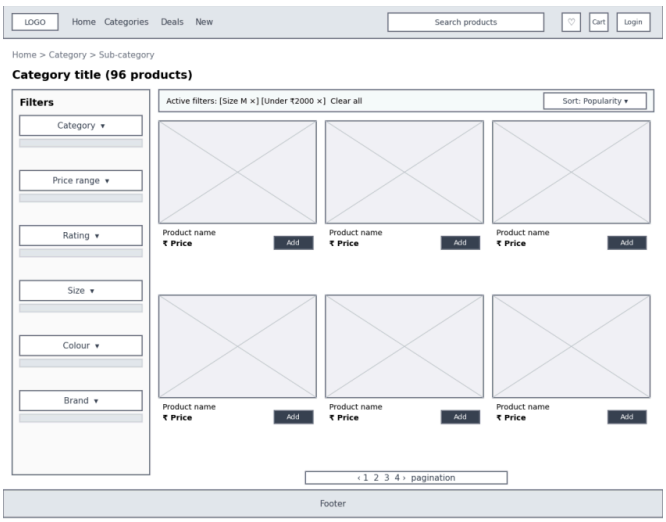


Figure 5.3: Low-fidelity wireframe of the product listing page with filter sidebar (illustrative).

The wireframe places filters in a left sidebar on desktop, a sort control above the grid, and active filter chips directly below the toolbar so that the current state of the list is always visible. Pagination is shown at the bottom; an alternative such as a “Load more” button could be considered, but whichever pattern is chosen should preserve the shopper’s place when they return from a product page.

5.8 Product Detail Page

The product detail page helps users evaluate a selected item. A typical layout includes product imagery, a product name, price, available variants, quantity selection, a primary action, and supporting details. Information should be grouped so that shoppers can compare the most important attributes without searching through an unstructured block of text.

The “Add to Cart” action should have an understandable label and a visible response. If a required option is not selected, the interface should identify the missing choice. Details such as delivery dates, stock availability, warranties, or return terms should only be displayed when accurate information is

available from the eventual system.

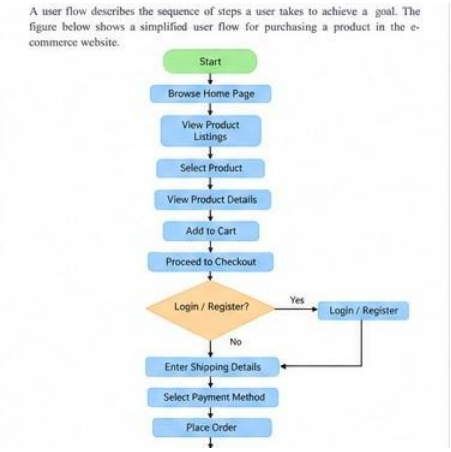


Figure 5.4: Simplified purchase flow from the home page to placing an order (illustrative). A high-fidelity product detail mockup is not included in this report.

A high-fidelity mockup of this page would demonstrate several design decisions. The price would be the largest text after the product name, with the original price struck through and the discount shown in a distinct label so that it can be recognised without reading. Colour options would be shown as swatches and size options as selectable chips, with a quantity stepper beside them. The primary action, “Add to Cart”, would use the strongest colour on the page, while “Buy Now” would use an outlined style so that the two actions are visibly different in importance. Tabs below the fold would group description, specifications and reviews. The wireframe in Figure 5.5 abstracts this layout.

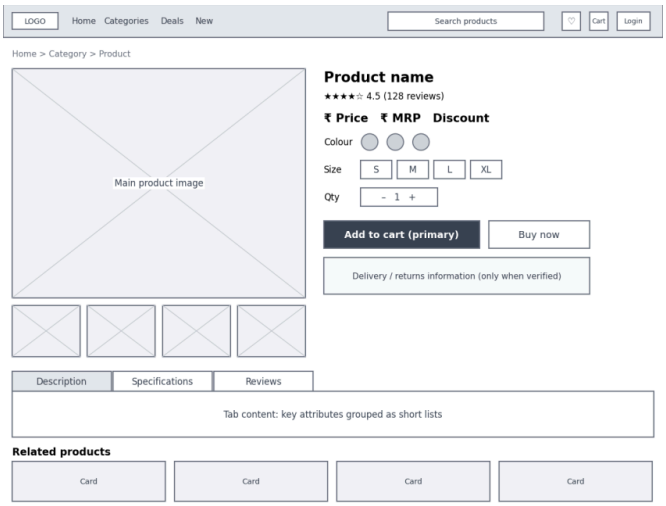


Figure 5.5: Low-fidelity wireframe of the product detail page (illustrative).

5.9 Shopping Cart

The cart is a review point before checkout. It should identify each selected product, option, quantity, unit price, and line total where applicable. Users should be able to change quantities or remove items, and the interface should make the updated state understandable. A summary area can show subtotal and

any known charges, but unknown shipping or tax values should not be invented.

The cart should also provide a clear route back to shopping and a prominent next step toward checkout. Empty-cart and unavailable-item states are important design cases even if the illustrative report does not depict every state.

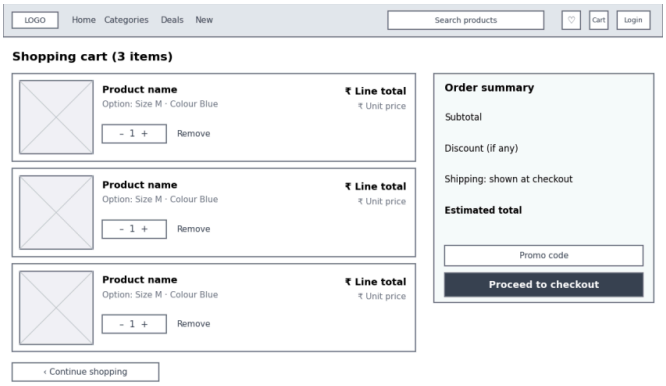


Figure 5.6: Low-fidelity wireframe of the shopping cart (illustrative).

In the wireframe the order summary is placed to the right of the item list so that the shopper sees the effect of quantity changes immediately. The shipping line is shown as “calculated at checkout” rather than as a number, reflecting the principle that the interface should not display values that the system cannot yet determine. On mobile the summary would stack below the items, with the checkout button repeated in a sticky bar.

5.10 Checkout Interface

Checkout is a multi-step interaction that may collect contact, delivery, and payment information depending on the eventual service. The interface should clearly identify required fields, group related information, explain validation errors, and allow review before an order is placed. Sensitive payment information should be handled only through an appropriately secure implementation; a visual mockup does not establish payment security.

A checkout design should minimize unnecessary re-entry and explain what happens after each step. Progress indicators may help when checkout spans multiple stages. The final confirmation screen should communicate order status only after a real system has confirmed the transaction. In this report, checkout and confirmation are proposed screen concepts.

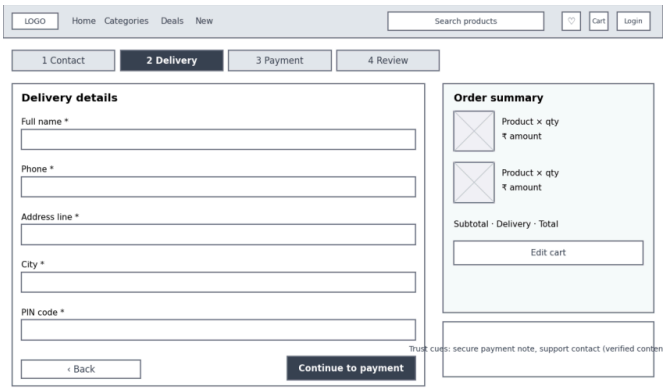


Figure 5.7: Low-fidelity wireframe of the delivery step in checkout (illustrative).

5.11 Login and Registration

If an account is part of the intended store, login and registration screens should use clear labels, distinguish sign-in from account creation, and provide understandable error and recovery states. Password rules should be explained before submission. A guest-checkout option may be considered if consistent with the product requirements.

Authentication behaviour, password storage, account security, and identity verification belong to the implementation layer and are outside this design report. The interface can describe states and interactions, but it cannot by itself provide the underlying security functions.

5.12 Order Confirmation and Post-Purchase States

The confirmation screen closes the purchase journey. In a real system it would be displayed only after the transaction has been confirmed, and it would summarize the order, state what happens next and offer a route back to shopping. Because this report documents design concepts only, the confirmation screen is described but not claimed as a functioning feature.

5.13 Empty, Error and Loading States

Designs are often presented only in their ideal state, but real users meet empty lists, slow connections and failed actions. Planning these states early prevents the interface from feeling unfinished when something goes wrong. Table 5.3 lists proposed treatments for common states in the illustrative store.

Table 5.3: Proposed treatment of empty, error and loading states

State	Where	Proposed treatment
Empty cart	Cart	Friendly message, “Browse categories” button, optional saved items
No search results	Search results	Restate the query, suggest checking spelling, offer popular categories
No products match filters	Listing	State that no products match, show active filters, offer “Clear filters”

State	Where	Proposed treatment
Loading product grid	Listing	Skeleton cards with the same layout as final cards
Option not selected	Detail	Inline message beside the missing option; keep other choices intact
Item unavailable	Cart	Flag the item, explain the issue, offer remove or similar items
Form validation error	Checkout / Account	Message next to field, summary at top for several errors, preserve entered data
Network failure	Any	Plain-language message with a retry action; avoid showing technical codes

5.14 Mobile Layout Concepts

A mobile shopping experience should prioritize the task rather than preserve every desktop element. A compact header, accessible search, readable product cards, clear action buttons, and a stacked product detail layout can support smaller screens. Fixed controls should not obscure content or form fields.

Mobile design should also account for long product names, variable image proportions, larger text settings, keyboard display, and scrolling. The layout should be reviewed at several widths and with realistic content. The visual shown in this report is an illustrative concept and is not a record of a completed responsive prototype.

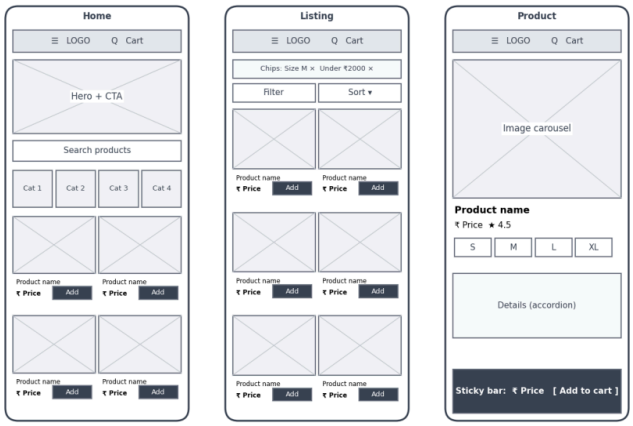


Figure 5.8: Wireframe concepts for mobile home, listing and product detail screens (illustrative).

5.15 Visual Design System

A visual design system documents the colours, type, spacing and components used across the interface so that screens remain consistent as the design grows [9]. Table 5.4 summarizes the proposed guidelines. Following the contrast calculations in Chapter 3, the proposed system uses a deeper accent value for text-bearing buttons.

Table 5.4: Proposed visual design system guidelines

Element	Proposed guideline
Primary colour	Navy (#0B1F4B) and blue (#1E4FD8) for structure and key actions; verify contrast before implementation
Accent	#BE185D for text-bearing buttons (about 6.0:1 on white); brighter pink only for large graphics where 3:1 is sufficient
Background	Light neutral surfaces (#F5F7FB) to separate content areas
Typography	Limited, consistent type scale for headings, body text and metadata
Spacing	Reusable spacing increments (4, 8, 16, 24, 32, 48 px) and consistent grid alignment
Buttons	Distinct primary and secondary action hierarchy
Product cards	Consistent image, name, price and action placement
Form fields	Persistent labels, visible focus, and clear error messages
Status colours	Success and error colours always accompanied by text or icon
States	Document default, focus, disabled, selected, loading and error states as applicable

5.16 Iconography, Imagery and Content Guidelines

Icons should be drawn from one family with a consistent stroke weight and style, and should be paired with text labels wherever the meaning is not universally understood. Icons that stand alone, such as the cart or wishlist heart, need accessible names for assistive technology.

5.17 Figma File Organization

A maintainable Figma file could be organized into pages such as “Foundations”, “Components”, “Wireframes”, “Desktop Screens”, “Mobile Screens”, and “Prototype” [1]. Frames should use meaningful names and a predictable grouping structure. Components can be used for repeated controls and product cards, while styles or variables can document the colour and typography system.

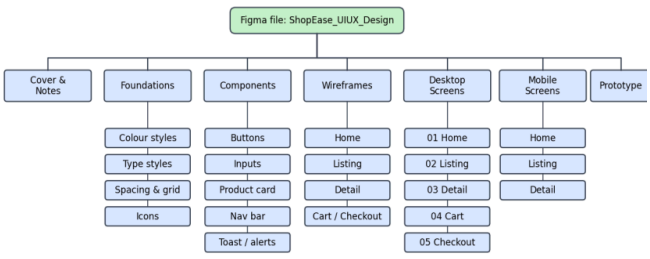


Figure 5.9: Recommended Figma file organization (not an original project file).

This is a recommended organization, not a claim that the student created these exact Figma pages. The supplied materials contain no Figma file or prototype

link, so the report does not include an invented URL.

5.18 Prototype Interaction Plan

Table 5.5: Prototype interaction plan

Trigger	Expected design response	Review question
Select a category	Open the relevant listing concept	Is the destination obvious?
Choose a product card	Open the matching detail concept	Does the product identity remain clear?
Add to cart	Show visible confirmation and cart count/state	Can the user tell the action succeeded?
Edit cart quantity	Update displayed quantity and totals in a real implementation	Is the change easy to understand?
Proceed to checkout	Open checkout step or sign-in branch	Can users recover and return?
Place order	Show confirmation only after system confirmation	Does the message avoid false success?

In a Figma prototype these triggers would be implemented as connections between frames or between component variants, using on-click interactions and simple transitions such as dissolve or slide [1]. The prototype could be shared by link for review, but it would remain a demonstration of sequence and clarity, not a working shop.

5.19 Design Review Checklist

A design review should check consistency across screens, alignment to a shared grid, readable type, sufficient contrast, clear action hierarchy, sensible content grouping, and meaningful feedback. It should also examine empty, error, loading, disabled, and selected states where relevant. Responsive screens should be checked for wrapping, overlap, clipping, and touch-target spacing.

No formal test results are reported here. The checklist is a proposed review aid that could be used in a later validation stage. Actual usability claims would require documented tasks, participant or evaluator observations, and a record of issues and revisions.

5.20 Design Risks and Mitigations

Table 5.6: Design risks and mitigations

Risk	Possible mitigation
Crowded product pages	Prioritize essential information and group secondary details
Inconsistent controls	Use shared components and visual rules
Unclear purchase status	Specify visible confirmation and error states
Poor mobile fit	Review responsive frames at multiple widths

Risk	Possible mitigation
Insufficient contrast	Check text and control contrast against WCAG guidance [2]
Unsupported business details	Use placeholders until verified requirements are available
Long or variable content breaking layouts	Test cards and forms with realistic long names and large text settings
Dark patterns or misleading urgency	Avoid false scarcity messages; show only accurate stock and delivery information

6. INTERNSHIP ACTIVITY SCHEDULE AND LEARNING OUTCOMES

6.1 Activity Schedule Overview

The supplied activity categories are organized into three phases over 30 days. This schedule is a structured summary of the stated learning plan, not a time-stamped attendance log. It does not invent individual mentor meetings, user interviews, client reviews, or formal test outcomes. The day-wise entries are planned learning topics; the internship activities documented by the organization are limited to the facts recorded in the supplied documents (Appendix A). Figure 6.1 presents the schedule graphically, and Table 6.1 lists the learning topic for each day.

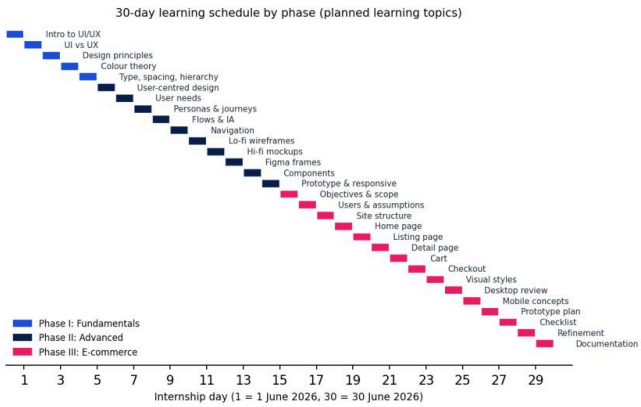


Figure 6.1: Learning schedule by phase and day (planned learning topics, not an attendance log).

Table 6.1: 30-day activity schedule

Day	Learning / design activity	Phase
1	Introduction to UI and UX	Phase I – Fundamentals
2	Differences between UI and UX	Phase I – Fundamentals
3	Principles of effective interface design	Phase I – Fundamentals
4	Colour theory and palette selection	Phase I – Fundamentals
5	Typography, spacing, alignment and visual hierarchy	Phase I – Fundamentals

Day	Learning / design activity	Phase
6	User-centred design principles	Phase II – Advanced UI/UX
7	Understanding user needs and requirements	Phase II – Advanced UI/UX
8	User personas and user journeys (assumption-based)	Phase II – Advanced UI/UX
9	User flows and information architecture	Phase II – Advanced UI/UX
10	Navigation and content organization	Phase II – Advanced UI/UX
11	Low-fidelity wireframes	Phase II – Advanced UI/UX
12	High-fidelity mockup concepts	Phase II – Advanced UI/UX
13	Figma frames and layout organization	Phase II – Advanced UI/UX
14	Reusable components and design consistency	Phase II – Advanced UI/UX
15	Prototyping, responsive design and usability considerations	Phase II – Advanced UI/UX
16	Project objectives and scope	Phase III – E-commerce project
17	Intended users and design assumptions	Phase III – E-commerce project
18	Website structure and navigation	Phase III – E-commerce project
19	Home page wireframe and layout	Phase III – E-commerce project
20	Product listing page design	Phase III – E-commerce project
21	Product detail page design	Phase III – E-commerce project
22	Shopping cart interface planning	Phase III – E-commerce project
23	Checkout interface planning	Phase III – E-commerce project
24	Visual styles and reusable components	Phase III – E-commerce project
25	Desktop layout review	Phase III – E-commerce project
26	Mobile layout concepts	Phase III – E-commerce project
27	Prototype interaction planning	Phase III – E-commerce project
28	Consistency and usability checklist	Phase III – E-commerce project
29	Refinement opportunities and limitations	Phase III – E-commerce project
30	Project documentation and final summary	Phase III – E-commerce project

6.2 Phase I: UI/UX Fundamentals (Days 1–5)

The first phase introduces the language and principles of interface design. UI and UX are considered together, with attention to how layout, colour, typography, and spacing can make a digital product understandable. Basic wireframing is introduced as a way to plan page structure before detailed styling.

The learning outcome of this phase is a vocabulary for discussing design decisions. For example, a product card can be evaluated for visual hierarchy, readable price presentation, consistent spacing, and whether its action is clearly identifiable.

6.3 Phase II: Advanced UI/UX Design (Days 6–15)

The second phase expands from individual visual elements to the full user journey. User-centred design provides a framework for considering intended users and their tasks. Personas and journeys are treated as planning artefacts based on assumptions unless real research evidence becomes available. Information architecture and user flows help organize screens and show how users move through the experience.

Wireframes are used to plan structure, while high-fidelity mockups explore the visual system. Figma frames, reusable components, and consistent styles are relevant techniques for keeping related screens aligned. Prototyping can demonstrate intended navigation and interaction states. Responsive design introduces the need to adapt content and controls to different screen sizes.

6.4 Phase III: E-Commerce Project (Days 16–30)

The final phase applies the earlier concepts to a proposed e-commerce experience. The project begins with scope and user assumptions, then establishes a site structure and a coherent screen set. Home, listing, detail, cart, and checkout concepts are documented as parts of a single purchase journey.

The design review focuses on consistency, readability, navigation, action hierarchy, and responsive considerations. The final documentation identifies what the mockups communicate and what remains outside the design scope, such as functional implementation, formal usability testing, and integration with real commerce services.

6.5 Skills and Learning Outcomes

Table 6.2: Learning outcomes and evidence status

Learning area	Outcome described in this report	Evidence status
UI/UX fundamentals	Ability to explain core visual and experience principles	Learning category supplied; no separate assessment record provided

Learning area	Outcome described in this report	Evidence status
Information architecture	Ability to organize a proposed shopping structure	Documented as a design concept
Wireframes and mockups	Understanding of structure-first and visual-refinement stages	Illustrative mockups and wireframes included
Figma concepts	Understanding of frames, components and prototypes	No original Figma file supplied
Responsive design	Awareness of desktop/mobile layout differences	Conceptual discussion included
Usability and accessibility	Awareness of review questions and WCAG relevance	No formal evaluation claimed
Documentation	Ability to record design intent, assumptions and limitations	This report

6.6 Challenges and Design Considerations

Common challenges for an e-commerce design include presenting enough product information without crowding the page, keeping actions consistent across screens, and adapting navigation to mobile layouts. Another consideration is communicating state: users need to understand when an item is selected, added to the cart, or awaiting required information.

These are general design considerations, not a claim that each issue was observed during the internship. A future project record could distinguish actual problems encountered, feedback received, revisions made, and decisions accepted by a supervisor. No such detailed evidence was supplied for this report.

6.7 Professional Practice and Documentation

Design documentation should communicate the intent behind each screen and the rules that keep the interface consistent. Clear names, reusable components, organized frames, and a documented user flow make a design easier for another person to review. Captions and notes should distinguish a visual concept from a functional requirement.

A handoff package may include screen dimensions, component variants, typography and colour specifications, responsive behaviour, and notes for empty or error states. If the project proceeds to development, designers and developers should clarify which behaviours are required and how they will be verified.

6.8 Reflection

The internship topic offered a context for connecting design theory with a proposed shopping interface. The report emphasizes the importance of organizing user tasks before styling individual screens and of using a consistent visual language across the journey. It also highlights the limits of

mockups: a clear visual concept still requires implementation, accessibility review, and user validation before it can be treated as a complete digital service.

7. CONCLUSION, LIMITATIONS AND FUTURE ENHANCEMENTS

7.1 Conclusion

This report documents the UI/UX design learning associated with the 30-day internship at Astonish InfoTech Private Limited from 1 June 2026 to 30 June 2026. The supplied certificate identifies the internship topic as UI/UX Design, and the annexure records the project work as E-Commerce. The report organizes the project as a proposed e-commerce website interface design using Figma-related concepts.

The main design principles discussed include clarity, consistency, visual hierarchy, readable typography, meaningful navigation, reusable components, interaction feedback, and responsive adaptation. The proposed screen set demonstrates how these principles can be considered across product discovery and purchase-related steps. The supporting analyses, such as the contrast calculation, the experience requirements, the state treatments and the review criteria, show how a design can be reasoned about before it is built.

7.2 Limitations

The report is based on the supplied internship certificate and the user-provided learning schedule. No original Figma file, authenticated project screenshots, source code, test records, user-research notes, or deployment evidence were supplied. Accordingly, the interface images are labelled illustrative, and no claim is made that a production website, backend, database, payment flow, or live prototype was implemented.

The personas and user requirements are assumptions used to explain the design process. No formal usability testing, accessibility audit, measured performance improvement, or client acceptance is claimed. The unsigned bonafide certificate is an academic template and must be completed by authorized college personnel if required.

7.3 Future Enhancements

Future work could include collecting requirements from representative users, validating personas through research, conducting moderated usability tasks, and documenting observed issues and revisions. The design could be expanded with additional responsive breakpoints, keyboard interaction states, screen-reader labels, and a formal WCAG accessibility review [2].

If the design is selected for implementation, frontend development, backend and database integration, inventory and order management, secure

authentication, and payment-provider integration would be separate engineering activities. They should be implemented and tested by qualified personnel; their presence is not implied by this UI/UX report.

7.4 Final Summary

The internship report presents a structured account of UI/UX learning and a proposed e-commerce design case study. It records the official internship dates and organization details from the supplied documents, describes a coherent 30-day learning schedule, and provides illustrative screens to make design decisions easier to understand. It also states evidence limitations clearly so that proposed concepts are not mistaken for verified implementation outcomes.

8. REFERENCES

[1] Figma. (n.d.). *Figma Help Center*. https://help.figma.com/ (Accessed 12 October 2026).

[2] World Wide Web Consortium (W3C). (2023). *Web Content Accessibility Guidelines (WCAG) 2.2*. https://www.w3.org/TR/WCAG22/

[3] World Wide Web Consortium (W3C). (n.d.). *WAI tutorials*. https://www.w3.org/WAI/tutorials/

[4] Nielsen Norman Group. (n.d.). *Articles on usability, user experience, and interaction design*. https://www.nngroup.com/articles/

[5] Norman, D. A. (2013). *The Design of Everyday Things: Revised and Expanded Edition*. Basic Books.

[6] Krug, S. (2014). *Don't Make Me Think, Revisited: A Common Sense Approach to Web Usability* (3rd ed.). New Riders.

[7] International Organization for Standardization. (2019). *ISO 9241-210:2019 Ergonomics of human-system interaction—Part 210: Human-centred design for interactive systems*. ISO.

[8] Garrett, J. J. (2011). *The Elements of User Experience: User-Centered Design for the Web and Beyond* (2nd ed.). New Riders.

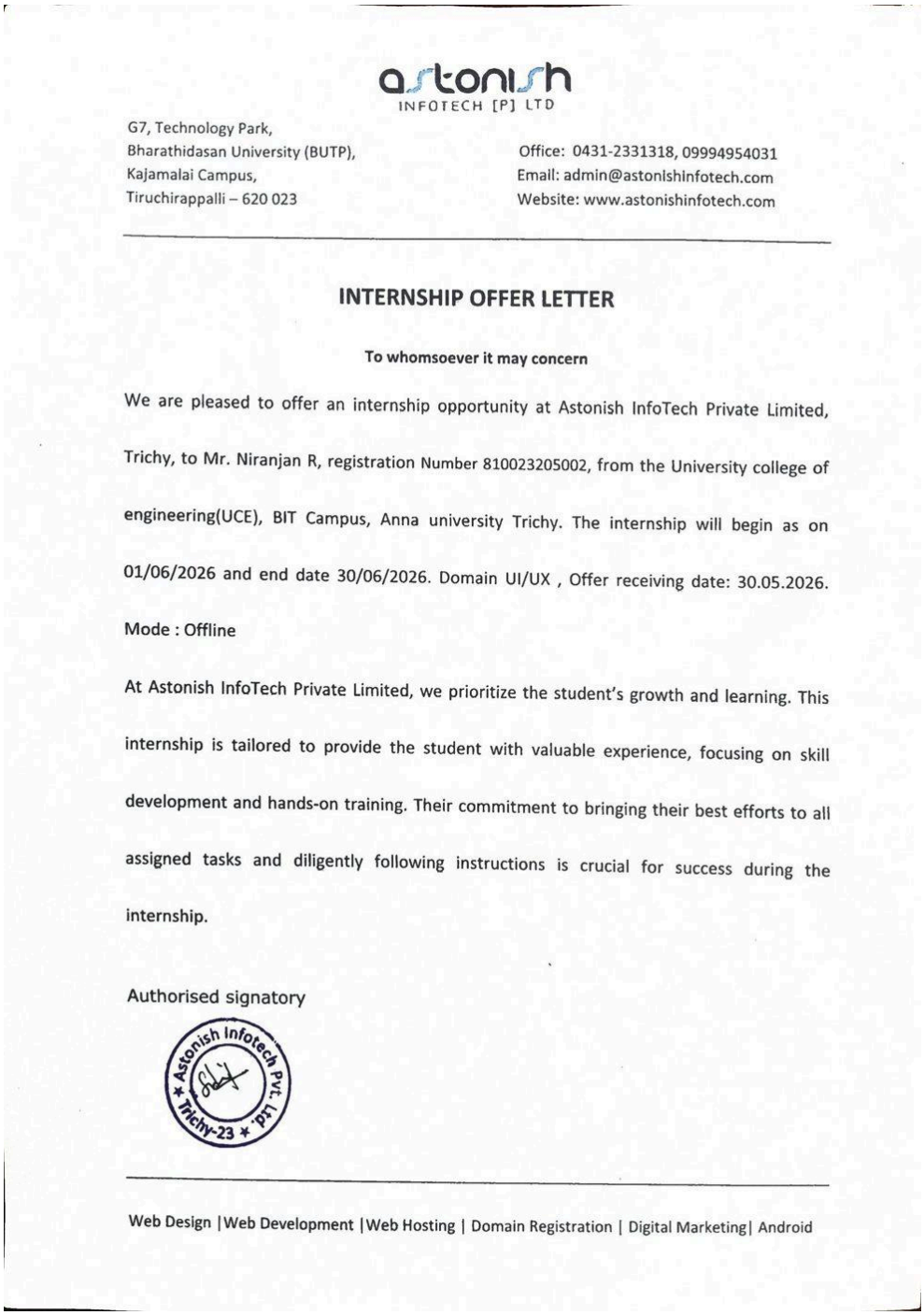
[9] Frost, B. (2016). *Atomic Design*. Brad Frost.

[10] Astonish InfoTech Private Limited. (2026). Internship offer letter, internship completion certificate, attendance certificate and annexure supplied by the student.

[11] Nielsen, J. (1994). *10 usability heuristics for user interface design*. Nielsen Norman Group. https://www.nngroup.com/articles/ten-usability-heuristics/

APPENDIX A – INTERNSHIP DOCUMENTS

Page 1 is the internship offer letter; page 2 is the internship completion certificate; page 3 is the attendance certificate; and page 4 is the internship annexure.



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INTERNSHIP OFFER LETTER

To whomsoever it may concern

We are pleased to offer an internship opportunity at Astonish InfoTech Private Limited, Trichy, to Mr. Niranjan R, registration Number 810023205002, from the University college of engineering(UCE), BIT Campus, Anna university Trichy. The internship will begin as on 01/06/2026 and end date 30/06/2026. Domain UI/UX , Offer receiving date: 30.05.2026.

Mode : Offline

At Astonish InfoTech Private Limited, we prioritize the student's growth and learning. This internship is tailored to provide the student with valuable experience, focusing on skill development and hands-on training. Their commitment to bringing their best efforts to all assigned tasks and diligently following instructions is crucial for success during the internship.

Authorised signatory

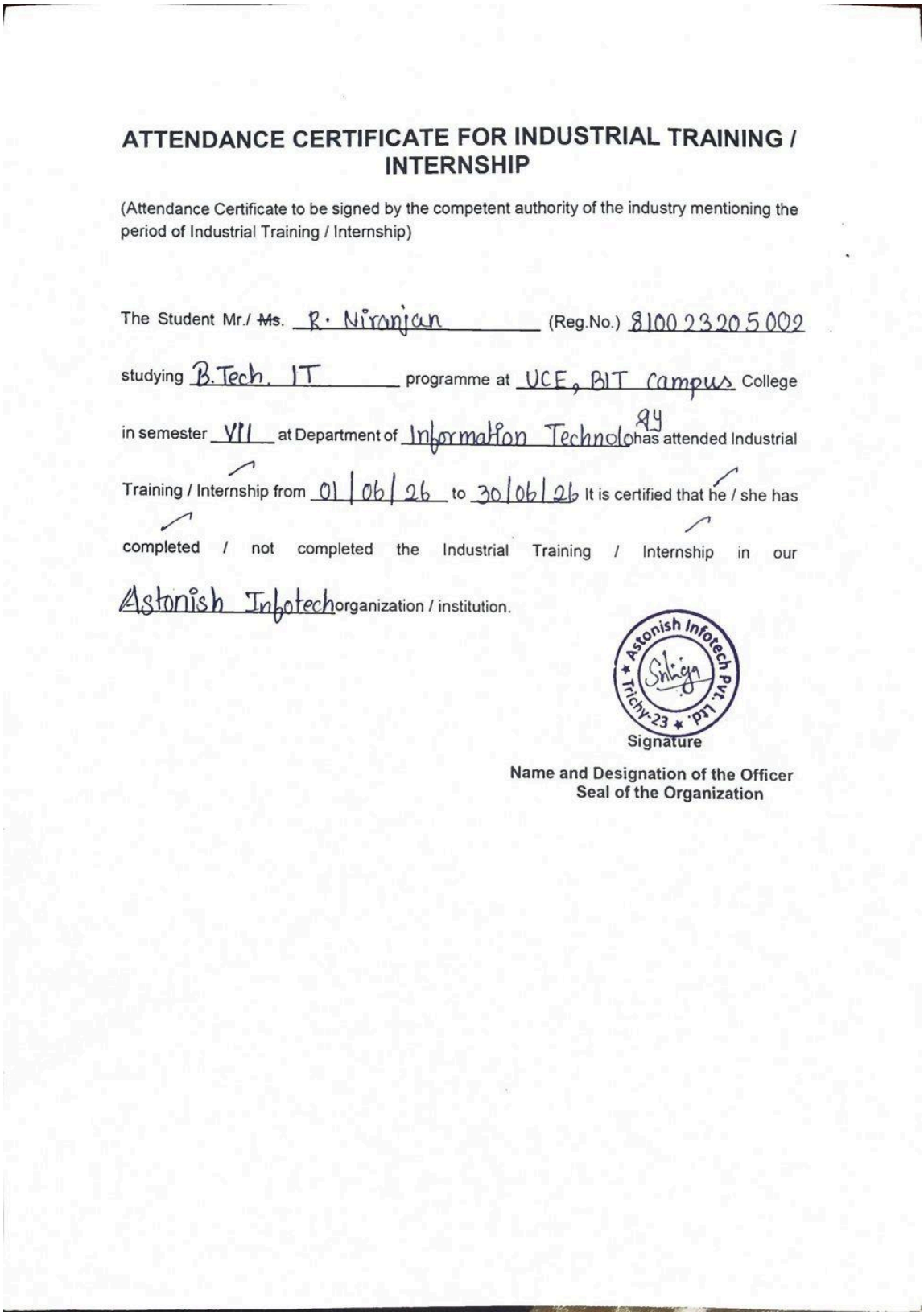
Astonish Infotech Pvt. Ltd. Trichy-23

Web Design |Web Development |Web Hosting | Domain Registration | Digital Marketing| Android

Figure A.1: Internship offer letter (original scan).



Figure A.2: Internship completion certificate (original scan).



ATTENDANCE CERTIFICATE FOR INDUSTRIAL TRAINING / INTERNSHIP

(Attendance Certificate to be signed by the competent authority of the industry mentioning the period of Industrial Training / Internship)

The Student Mr./ ~~Ms~~. R. Niranjan (Reg.No.) 810023205002

studying B.Tech. IT programme at UCE, BIT campus College

in semester VII at Department of Information Technology has attended Industrial

Training / Internship from 01/06/26 to 30/06/26 It is certified that he / she has

completed / not completed the Industrial Training / Internship in our

Astonish Infotech organization / institution.

Astonish Infotech Pvt. Ltd. Trichy-23

Signature

Name and Designation of the Officer
Seal of the Organization

Figure A.3: Attendance certificate for industrial training / internship (original scan).

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Email: admin@astonishinfotech.com
Website: www.astonishinfotech.com

ANNEXURE

To Internship (June-July2026)

1.Name of the Industry with address: Astonish Infotech, G7, Technology Park, Trichy.

2.Name of the Industrial Expert (Joint Supervisor): Srikanniga Priyadharshini. R

3.Qualification of the Industrial Expert (Joint Supervisor): B.Tech-IT (Hons)

4.Designation of the Industrial Expert (Joint Supervisor): Project Manager

5.Title of the Project Work: E-Commerce

6.Duration of the Internship: 1 Month

7.Details of the student:

SI.No	Name of the Student	Registration No	Department
1.	R. Niranjan	810023205002	IT

Terms and conditions

- The student is permitted to undergo a semester long internship in your organization
- The student shall be instructed to meet the Institute supervisor periodically and to attend the review committee meetings for evaluating the progress.
- The final report of the internship duly signed by the industrial expert/supervisor will be submitted to college.
- Attendance Certificate in the prescribed format signed by the competent authority of the industry will be produced.

Authorised signatory

Astonish Infotech Pvt. Ltd. Trichy-23

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Figure A.4: Internship annexure (original scan).